UNIT-1
WEB CONTENT MANAGEMENT SYSTEM

Web Content Management System (WCMS)

Definition: *A Web Content Management System (WCMS) is a software system that enables users to create, manage and publish content on web pages, without requiring programming knowledge.*

A Web Content Management System (WCMS) allows users to create, manage and deploy content on web pages easily. It removes the need for coding knowledge to publish content on the web.

What a WCMS Manages

- Text
- Graphics and Photos
- Audio
- Video
- Web pages and other digital content

Popular Examples

- WordPress
- Joomla
- Drupal

These systems make web publishing faster, easier and more affordable.

Features / Characteristics of a WCMS

- **Easy Content Creation** — A WCMS allows users to create and edit website content such as text, images, audio and video easily, without requiring extensive programming knowledge.
- **Template and Theme Support** — It provides ready-made templates and themes that control the appearance of a website. This helps maintain a **consistent design and layout** across different webpages.
- **User Roles and Permissions** — A WCMS allows different users to have different levels of access, such as **administrator, editor, author and viewer**, according to their responsibilities.

RAKSHITA

- **Workflow Management** — It provides a systematic process for handling content. Content can be **created, reviewed, approved and then published**, helping to maintain quality and accuracy.
- **Media Management** — A WCMS provides a central media library to **store, organize and reuse** images, audio, video and other media files.
- **Multi-user Collaboration** — Multiple users can work on website content according to their assigned roles. This makes it easier for a team to **create, edit and manage content together**.
- **Version Control** — A WCMS keeps track of changes made to content. It may allow users to **view previous versions and restore an earlier version** when required.
- **SEO (Search Engine Optimization) Tools** — A WCMS may provide tools for adding **page titles, keywords, descriptions and metadata** to help improve the visibility of website content in search engine results.
- **Integration** — A WCMS can **connect with other software, platforms and online services** to exchange data and provide additional functionality. For example, a college website can integrate a **payment gateway, email service, social media platform or analytics tool**.
- **Extensibility** — A WCMS can be extended by adding **plugins, extensions or modules**. These add new functions and capabilities to the basic system.
- **Security** — A WCMS provides features such as **user authentication, access control and permissions** to protect website content from unauthorized access or modification.

Web Content

Definition: *Web Content is defined as the collection of information and media — such as text, images, audio, video, documents and interactive elements — that is displayed or provided on a website for its users.*

Examples of Web Content

- **Text —** Articles, headings, descriptions, news
- **Images —** Photos, logos, diagrams, graphics
- **Audio —** Music, podcasts, voice recordings
- **Video —** Tutorials, lectures, advertisements
- **Documents —** PDF files, reports, forms
- **Links —** Links to other webpages/websites
- **Interactive Content —** Forms, quizzes, buttons, surveys

Example: *A college website contains — college information, notices, photos, videos, timetable and admission forms. All of these are web content.*

Characteristics of Good Web Content

- **Informative** — Gives useful and meaningful information to the user
- **Accurate** — Free from errors and factually correct
- **User-Focused** — Written keeping the target audience's needs in mind
- **Well-Structured** — Organized under clear headings and sections
- **Engaging** — Holds the reader's attention through relevant text/media
- **Multimedia-Rich** — Uses text, images, audio and video where suitable
- **Regularly Updated** — Old/outdated information is refreshed on time
- **Accessible** — Can be read and used by all types of users and devices

Web Content Development

Definition: *Web Content Development is defined as the process of planning, creating, editing and preparing content for publication on the web.*

In simple terms, it means creating useful and suitable content for a website.

Tip: *Web Content Development = Planning + Creating + Editing + Organizing + Publishing Content.*

Web Content Development Includes

- Planning the required content
- Creating text, images, audio and video
- Editing and improving the content
- Organizing the content properly
- Formatting the content for the web
- Publishing the content on a website
- Updating the content when required

Example: *For a college website: college & course information, admission details, notices, event information, images/videos and online forms all need to be developed.*

Web Content Management

Definition: *Web Content Management is defined as the process of creating, editing, organizing, publishing, updating and maintaining content on a website after it has been created.*

RAKSHITA

Key Functions of Web Content Management

- **Create** — Allows users to **add new content** to a website. The content may include text, images, videos, documents, announcements or webpages.
  Example: Adding a new page containing BCA admission details.
- **Edit** — Enables users to **correct, modify or improve existing content**. Errors can be corrected and information can be changed whenever required.
  Example: Correcting a spelling mistake in a college notice.
- **Organize** — Helps to **arrange content in a proper and logical structure** so that users can find information easily. Content can be grouped into pages, categories or sections.
  Example: Organizing college information into Admissions, Courses, Examinations and Notices.
- **Publish** — Makes completed content **available to website users**. Content can be reviewed and approved before being published.
  Example: Publishing an examination timetable on the college website.
- **Update** — Enables users to **replace outdated information with current information**. Regular updates ensure that users receive the latest information.
  Example: Changing an old admission deadline when the university announces a new date.
- **Maintain** — Helps to keep website content **accurate, useful, relevant and up to date** over time. It may involve checking information, correcting errors and removing outdated content.
  Example: Removing an old college notice after the event is completed.

Example: *On a college website, admission information, exam timetable, notices, events, faculty details and photos/videos need regular management. If the admission date changes, the old date must be edited and the new date published.*

RAKSHITA

Development vs Management — Quick Comparison

Web Content Development	Web Content Management
Focuses on creating content	Focuses on managing content
Planning and preparing content	Editing and updating content
Creates new information	Maintains existing information
Example: Creating an admission page	Example: Updating admission dates

Importance of Web Content Development and Management

• **Provides Information** — Gives users useful and relevant information about a website, organization, product or service.
Example: A college website provides course and admission details.
• **Attracts Users** — Good and interesting content encourages users to visit and spend more time on a website.
Example: A website with useful articles, images and videos is more attractive.
• **Improves Understanding** — Clear and simple content helps users understand information easily.
Example: A step-by-step admission guide is easier to understand.
• **Supports Communication** — Helps organizations share information with students, customers, employees and the public.
Example: A college can publish notices and announcements on its website.
• **Builds Trust** — Accurate and regularly updated information makes a website more reliable and trustworthy.
Example: Providing correct examination dates increases students' confidence in the website.
• **Keeps Information Updated** — Regular management helps replace old information with the latest information.
Example: Updating the admission date when it changes.

• **Improves User Experience** — Well-organized content helps users find and use the required information easily.
Example: Clearly arranged menus such as **Admissions, Courses, Exams and Notices** make a college website easier to use.

PAKSHITA

Content Types

Definition: *A Content Type is defined as the different form in which information is presented to users on a website, such as text, images/graphics, audio, video, documents or interactive content.*

Example: *On a college website — description → Text, logo → Image, event recording → Video, announcement → Text, prospectus → Document, admission form → Interactive Content.*

Types of Web Content

1. Text and Written Information

Text is the **most common type of web content** used to provide information through written words.

Examples: Headings, paragraphs, descriptions, FAQs and instructions.

2. Images and Graphics

Images and graphics provide **visual information** and can make information easier to understand.

Examples: Photographs, logos, diagrams, illustrations and charts.

3. Audio

Audio is content presented through **sound or voice**. It is useful when information needs to be communicated through listening.

Examples: Podcasts, interviews, music and voice recordings.

4. Video

Video presents information through **moving images, usually along with sound**. It is useful for explaining or demonstrating information.

Examples: Tutorials, lectures, demonstrations and advertisements.

5. Documents

Documents are **digital files made available to users through a website**. They are commonly used to provide detailed or downloadable information.

Examples: PDF files, reports, brochures, notes and forms.

6. Interactive Content

RAKSHITA

Interactive content allows users to **interact with the website or provide information/input**.

Examples: Online forms, quizzes, surveys, polls and applications.

7. Articles

Articles are **detailed written content that provides information about a particular topic**.

Examples: Educational articles, technology articles and research articles.

8. Blog Posts

Blog posts are content **published regularly**, usually in an informative, educational or informal style.

Examples: Personal blogs, travel blogs and technology blogs.

9. News and Announcements

News and announcements provide **recent information, updates or important notices** to users.

Examples: College notices, event announcements and latest news.

10. Product or Service Information

This type of content **provides information about a product or service** to help users understand what is being offered.

Examples: Product descriptions, features, prices and service details.

Content Formats

Definition: *A Content Format is defined as the specific way in which digital content is stored, represented or presented, so that it can be shared and displayed properly on the web.*

Content Type = WHAT the content is stored/represented

Content Format = HOW the content is

Example: *Image → Content Type*

JPG → Content Format

LAKSHITA

Common Content Formats (Quick Table)

Content Type	Common Formats	Example
Text	TXT, DOC, PDF, HTML	Article, notes
Image	JPG/JPEG, PNG, GIF	Photo, logo
Audio	MP3, WAV	Music, podcast
Video	MP4, AVI, MOV	Tutorial, lecture
Document	PDF, DOCX	Report, brochure

1. Text Formats

TXT (Text File)

- Stores plain text without special formatting
- Simple letters, numbers and symbols only
- No fonts, images or advanced formatting

Example: *Simple notes, plain text → TXT → Plain text*

DOC / DOCX

- Formats for creating and storing formatted documents
- Support different fonts, sizes and styles
- Can include images, tables and other elements
- DOCX is the modern Microsoft Word format

Example: *Reports, assignments, letters → DOC/DOCX → Formatted document*

PDF (Portable Document Format)

- Preserves the layout of a document
- Can contain text, images, tables and other elements
- Commonly used for sharing/distributing documents

Example: *E-books, application forms → PDF → Fixed-layout document*

HTML (Hyper Text Markup Language)

- Used to create and structure web page content
- Contains headings, paragraphs, images, links etc.
- Web browsers interpret HTML to display web pages

Example: *Web pages, websites → HTML → Web page content*

2. Image Formats

JPG / JPEG

- Common format for storing photographs and digital images

RAKSHITA

- Reduces file size; suitable for photos and colourful images
- Does NOT support transparent backgrounds

Example: *Website photos, banners → JPG → Photographs*

PNG (Portable Network Graphics)

- Commonly used for graphics and digital images
- Supports transparent backgrounds; good image quality
- Suitable for logos, icons and diagrams

Example: *Logos, icons → PNG → Graphics + Transparency*

GIF (Graphics Interchange Format)

- Used for simple graphics; supports simple animations
- Limited colours compared to JPG/PNG

Example: *Simple animated icons → GIF → Simple Graphics + Animation*

3. Audio Formats

MP3

- Commonly used digital audio format
- Reduces file size; easy to store and share
- Good audio quality with smaller file size

Example: *Music, podcasts → MP3 → Smaller file size + widely used*

WAV

- Stores high-quality digital sound
- Higher quality than MP3 but needs more storage
- Used in audio recording/editing

Example: *Voice recordings, sound effects → WAV → High quality + larger size*

4. Video Formats

MP4 (MPEG-4 Part 14)

- Common video format for storing/sharing video with audio
- Good quality with relatively small file size
- Widely supported by browsers and devices

Example: *Online lectures, tutorials → MP4 → Common web video format*

AVI (Audio Video Interleave)

- Stores audio and video together
- Good quality but larger file size than MP4

Example: *Recorded videos, video clips → AVI → Audio + Video*

MOV

- Developed by Apple; stores video, audio and other data
- Commonly used with Apple apps/devices

PAKSHITA

Example: *Recorded videos, movies → MOV → Apple multimedia format*

5. Document Formats

PDF

- Stores and shares documents while keeping original layout
- Appearance stays consistent across devices

Example: *Reports, e-books → PDF → Easy to share + preserves layout*

DOC / DOCX

- Used for editable documents with text formatting
- Content can be easily edited and updated

Example: *Assignments, resumes → DOC/DOCX → Editable + formatted document*

Importance of Choosing the Right Content Format

Choosing the correct content format helps information to be **stored, displayed, shared and used effectively**. The following factors should be considered when selecting a format:

- **1. File Size** — Different formats can produce different file sizes. A suitable format can help save storage space and make files easier to share.
  Example: MP3 generally requires less storage space than WAV.
- **2. Quality** — The format can affect the quality of the content. Some formats provide better quality but may require more storage space.
  Example: WAV is commonly used when higher audio quality is required.
- **3. Compatibility** — The selected format should be supported by the **device, software or web browser** being used.
  Example: MP4 is widely supported for playing videos on different devices and browsers.
- **4. Web Performance** — Suitable and reasonably sized files can help webpages **load more efficiently**.
  Example: JPG is commonly used for website photographs because it can reduce file size.
- **5. Purpose of Content** — The format should be selected according to **how the content will be used**.
  Example: PDF is suitable for sharing a document, while MP4 is suitable for sharing a video.

Easy Formula to Remember

Right Format = File Size + Quality + Compatibility + Web Performance + Purpose

RAKHITA

Tip: *Photograph → JPG || Logo with transparency → PNG || Music → MP3 || Web page → HTML | Document to share → PDF.*

Choose a format based on: Quality + File Size + Compatibility + Purpose.

Norms and Guidelines of Content Development

Definition: *Norms and Guidelines of Content Development are defined as the rules and recommended practices followed while creating web content, so that it is correct, clear, useful, easy to read and properly presented.*

Good Web Content = Clear + Accurate + Relevant + Readable + Well Organized + Consistent + Accessible

- **Clear** — Easy to understand
- **Accurate** — Contains correct information
- **Relevant** — Useful to the target users
- **Readable** — Easy to read and follow
- **Well Organized** — Information arranged properly
- **Consistent** — Similar style and presentation maintained
- **Accessible** — Usable by different types of users

Example: *A college exam notice should give the correct date, use a clear heading, present information in an organized way, avoid spelling/grammar errors and be easy for students to read.*

Important Guidelines

1. Defining Objectives and Strategy

- Define the purpose of the content
- Identify the target audience
- Understand users' needs and expectations
- Decide what information to provide
- Prepare a content plan

Example: *College website → admissions, courses and examination information.*

2. Content Creation and Quality

- Provide accurate and reliable information
- Use simple, clear language
- Keep content relevant and complete
- Use suitable text, images, audio and video
- Check spelling and grammar; use appropriate formats

Example: *Exam notice → correct date, time and venue.*

RAKSHITA

3. Accessibility and Inclusivity

- Make content easy to read
- Use suitable font size and contrast
- Provide alternative (alt) text for images
- Support keyboard navigation
- Make links/controls easy to identify
- Support different devices and platforms

Example: *Image → provide alternative text.*

4. Content Organization and Management

- Use clear titles and headings
- Group related information
- Use short paragraphs and lists
- Provide simple navigation
- Maintain a consistent layout; review regularly

Example: *College website → Admissions, Courses, Exams, Notices.*

5. SEO (Search Engine Optimization) and Discoverability

- Use meaningful page titles and relevant keywords
- Provide suitable descriptions and metadata
- Use descriptive links; add alt text to images
- Keep content useful and relevant

Example: *"BCA Admission 2026" is better than "Page 1".*

6. Maintenance and Evolution

- Review content regularly; update outdated information
- Correct errors; remove obsolete content
- Check links regularly
- Consider changing user needs
- Mention the last-updated date when required

Example: *Update the admission deadline whenever the date changes.*

Creating Digital Graphics

Definition: *Digital Graphics are defined as images or visual representations — such as pictures, drawings or visual designs — that are created, stored and displayed in digital form using computers or other electronic devices.*

Example: *A college website's logo, campus photographs, event posters and charts are all digital graphics.*

RAKSHITA

Examples & Uses in Web Content

Examples

- Photographs
- Logos
- Icons
- Diagrams
- Charts
- Digital drawings
- Illustrations

Uses in Web Content

- Make websites visually attractive
- Present information clearly
- Explain ideas using images and diagrams
- Improve overall presentation of web pages

Types of Digital Graphics

1. Raster Graphics — Characteristics

- Made of tiny dots called pixels
- Used mainly for photos and detailed images
- Resolution-dependent — quality is fixed to pixel count
- Enlarging too much makes the image blurred/pixelated
- Generally larger file sizes for high detail
- Examples: JPG, PNG, GIF

2. Vector Graphics — Characteristics

- Made using mathematical lines, curves and shapes
- Used mainly for logos, icons and simple illustrations
- Resolution-independent — can be enlarged without losing quality
- Usually smaller file size for simple designs
- Examples: AI, EPS

Tip: *Easy memory trick: Photo → Raster || Logo → Vector*

Basic Elements of Digital Graphics

- **Color** — Adds meaning and visual appeal
- **Line** — Creates direction and movement
- **Shape** — Used to create different objects/designs
- **Scale** — Controls the size of elements
- **Alignment** — Arranges elements in proper position
- **Space** — Empty area between design elements

AKSHITA

- **Contrast —** Makes important parts stand out

Steps in Creating Digital Graphics

Creating a digital graphic involves a series of steps to produce a **clear, attractive and suitable visual for its intended purpose**.

1. Plan the Graphic

First, decide the **purpose, target audience and message** of the graphic. This helps determine what information and design elements are required.

Example: Planning a poster to announce a college cultural event.

2. Create or Capture the Image

Create a new graphic using graphic design software or **capture an image using a camera or scanner**.

Example: Taking a photograph of a college event or creating a new logo.

3. Edit the Image

Modify the image to **improve its appearance and remove unwanted portions or errors**.

Example: Cropping an unnecessary part of a photograph or adjusting its brightness.

4. Add Design Elements

Add suitable **colours, shapes, lines, text, icons and other visual elements** to communicate the message effectively.

Example: Adding the event name, date, time and venue to a poster.

5. Set Size and Resolution

Choose the appropriate **size and resolution** according to where the graphic will be used.

Example: A graphic for a website may need a different size from one designed for printing.

6. Choose the File Format

Select a suitable file format based on the purpose of the graphic.

Examples: JPG, PNG and GIF.

7. Review the Graphic

RAKSHITA

Check the final design for **quality, clarity, alignment, appearance, spelling and other errors** before using it.

8. Save and Use

Save the completed graphic in the required format and use it on a **website, document, presentation, social media or other digital platform**.

> ***Example:*** *College event poster → Plan → Create/Capture images → Edit → Add text/colours/shapes → Set size → Save in format → Use on website.*

Digital Graphics Tools

- **Adobe Photoshop —** Image editing and graphic design; editing photos and creating designs
- **CorelDRAW —** Graphic design and illustrations; logos, posters, layouts
- **GIMP —** GNU Image Manipulation Program; open-source image editing tool

Common Features of Graphic Editing Tools

- Tools to create, edit, resize and crop images
- Support for layers, colours, filters and effects
- Text and shape tools for adding design elements
- Support for multiple file formats (JPG, PNG, GIF, etc.)
- Options to export/save graphics for web or print use

Uses of Digital Graphics

- **Websites —** Make web pages attractive and informative
- **Logos and Icons —** Represent organizations, brands, services
- **Advertisements —** Banners, posters, promotional designs
- **Education —** Explain concepts using diagrams/illustrations/charts
- **Presentations —** Make slides clearer and more engaging
- **Social Media —** Create posts, posters and visual content
- **Publications —** Brochures, magazines and digital documents

Basic Capabilities of Digital Graphics Tools

The commonly used tools can help users:

- Create digital graphics
- Edit images
- Resize images
- Improve image appearance
- Add design elements
- Prepare graphics for different uses

RAKSHITA

Audio Production and Editing

Definition: *Audio Production and Editing is defined as the process of recording, creating, modifying and preparing sound for use in digital content.*

In simple terms, Audio Production means recording sound, and Audio Editing means modifying and improving it to make it suitable for use.

Audio Can Include

- Voice recordings
- Music
- Sound effects
- Announcements
- Podcasts

Uses of Audio in Web Content

- To provide voice information
- To make content more engaging
- To explain information through audio
- To support videos and presentations
- To create podcasts and online learning content

Example: *Online college course → Record lecturer's voice → Remove unwanted parts → Improve sound → Save → Publish.*

Audio Production Process

Audio production is the process of **creating and preparing audio content for use in digital media and web content**. It involves recording, editing and preparing sound so that it can be used effectively.

Steps in Audio Production

1. Planning
Decide the **purpose, topic and type of audio** required before starting the production.

Example: Planning a podcast to provide information about a college event.

2. Recording
Record the required **voice, music or other sounds** using a microphone, mobile phone or other recording device.

Example: Recording the speaker's voice for the podcast.

RAKSHITA

3. Editing
Remove **mistakes, unwanted portions, unnecessary pauses and unwanted noise** from the recorded audio.

Example: Removing a wrong sentence from the recording.

4. Enhancing
Improve the audio by **adjusting its volume and sound quality** so that it is clearer and easier to listen to.

Example: Increasing the volume of a voice recording that is too low.

5. Mixing
Combine different audio elements such as **voice, music and sound effects** into a single audio track.

Example: Combining a speaker's voice with background music.

6. Saving and Exporting
Save the completed audio in a **suitable file format** so that it can be used or published as required.

Example: Saving a podcast as an MP3 file.

Simple Flow

Planning → Recording → Editing → Enhancing → Mixing → Saving & Exporting

> ***Example:*** *Podcast → Plan → Record voice → Remove mistakes → Adjust sound → Add music → Export.*

Audio Editing

Definition: *Audio Editing is defined as the process of manipulating sound recordings to improve their quality, remove unwanted sounds, or change the length or other properties of an audio clip.*

Four Main Steps

1. Selection

Select the **specific part of the audio** that needs to be edited.

Example: Selecting a portion containing a mistake or unwanted pause.

2. Editing

Make the **required changes** to the selected portion of the audio.

Example: Removing a mistake, cutting an unwanted section or correcting the length of the clip.

RAKSHITA

3. Processing

Apply suitable **effects or adjustments** to improve the audio.

Example: Adjusting the volume or applying an effect to improve the sound.

4. Output

Save the **final edited audio file** in a suitable format for use or publication.

Example: Saving the completed audio as an MP3 file.

Types of Audio Editing

1. Cutting

- Removes unwanted parts of a recording
- Used to remove mistakes, pauses or unnecessary sections

2. Fading

- Gradually increases or decreases volume
- Makes sudden volume changes smoother

3. Mixing

- Combines two or more audio tracks into one

Example: *Combining voice with background music.*

> **Tip:** *Podcast example → Cut mistakes → Fade beginning/end → Mix voice + background music.*

Simple Example: Creating a College Podcast

- **Plan** — Decide the topic and content.
- **Record** — Record the speaker's voice using a microphone or phone.
- **Select** — Choose the required parts of the recording.
- **Edit** — Remove mistakes, pauses and unwanted portions.
- **Process** — Adjust sound and apply necessary effects.
- **Mix** — Combine voice with suitable background music if required.
- **Output** — Save the final audio file.

> **Tip:** *Plan → Record → Select → Edit → Process → Mix → Output*

Characteristics of Good Quality Audio

- **Clarity** — Voice/sound is clear and easy to understand
- **Consistent Volume** — No sudden loud or very soft parts
- **Free of Noise** — No unwanted background noise or disturbance
- **Correct Format** — Saved in a format suitable for its purpose (e.g. MP3 for web)
- **Appropriate Duration** — Neither too long nor too short for the purpose
- **Well Synced** — Matches properly with video or presentation, if combined

AKSHITA